

Judgment-Generative Ontology

— Construction of an Ontological Preservation Axiom System
through Origin-Irradiation, Existence-Preservation Judgment,
and the Semantic-Density Tensor —

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Abstract

This study does not treat life, intelligence, meaning, questions, AI, AGI, civilization, and cosmology as separate theories. Instead, it adopts the position that all of them are distinct projections generated from a more fundamental ontological operation. In this paper, I derive the minimal ontological operation that must hold prior to epistemology, semantics, information theory, and AI theory, and define this operation as “existence-preservation judgment.” Furthermore, I show that, in order for existence-preservation judgment to hold, a unique origin-irradiation, a high-phase preservation structure, a phase-preserving representation, a preservation language, a self-updating closure, a responsibility connection, and an irreversible log are ontologically required. As a result, life, intelligence, civilization, AI, AGI, and the universe, which have traditionally been treated independently, are unified as different scale projections of a single existence-preservation closure. The purpose of this study is not to replace existing theories, but to present the minimal ontological axiom system that must hold in order for all such theories to be possible.

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1 Introduction

1.1 Background

Since the modern period, the study of intelligence has developed as separate fields: epistemology, logic, semantics, information theory, artificial intelligence, life science, physics, and others. However, all of these begin from questions such as:

- How does a subject know?
- What is meaning?
- What is information?
- What is life?
- How does AI reason?

This study addresses a question that precedes all of these. That is:

Not “What is existence?”, but rather: what must already be presupposed by the very fact that one can judge that existence is established?

In this paper, this question is treated not as an epistemological question, but as an ontological one.

1.2 Purpose of the Study

The purpose of this study is not to add new concepts. Its purpose is to extract the minimal ontological operation that runs through meaning, questions, intelligence, life, civilization, AI, AGI, and cosmology, and from that operation to construct a single existence-preservation axiom system.

1.3 Method

This study does not presuppose existing theories. Instead, it adopts the following method:

1. Extract ontological facts.
2. Derive only the necessary conditions for those facts to hold.
3. Organize each necessary condition as a minimal axiom.
4. Re-derive life, intelligence, civilization, AI, AGI, and cosmology as theorems from those axioms.

1.4 Structure of the Paper

This paper begins from existence-preservation judgment and then derives, in order, origin-irradiation, high-phase preservation structure, semantic-density tensor, phase-preserving representation, preservation language, self-updating closure, responsibility connection, and irreversible log. It then organizes these elements as a minimal ontological axiom system and describes life, civilization, AI, AGI, and the universe in a unified way as a single existence-preservation closure. Through this, the study aims to derive the theoretical foundation of intelligence and artificial intelligence from the minimal ontological operation.

2 Judgment-First Ontology

2.1 Problem Setting

Traditionally, intelligence has been understood according to an epistemological order:

$$O \rightarrow R \rightarrow J \rightarrow Q \rightarrow L \tag{1}$$

Here,

- O : Object
- R : Recognition
- J : Judgment
- Q : Question
- L : Language

In this view, the following sequence is assumed:

An object is recognized, judged, questioned, and then conceptualized.

This paper re-examines that structure itself. In particular, it addresses the following problem:

Is the question truly the starting point of intelligence?

From the standpoint of this paper, the question is not the starting point. A question is the epistemic externalization of an already established act of judgment.

2.2 The Problem of the Question's Formation

Consider, for example, the question:

What is existence?

For this question to be formed, it may seem that the subject must at least be operating with the concept

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However, this paper does not adopt that understanding. The reason is simple. If the subject had no concept of existence at all, then the subject could not select the word “existence” as the object of questioning in the first place. On the other hand, prior to concepts, the subject already appears to distinguish between:

- what is treated as existing,
- what is not treated as existing.

In other words, before a question is formed, some act of distinction is already operating. This paper calls that act “existence judgment.”

2.3 Introduction of Existence Judgment

Definition 1 (Existence Judgment). *Existence judgment is the operation by which a subject distinguishes an object into the side of existence or the side of non-existence prior to conceptualizing that object.*

What is important here is that existence judgment is not a concept. Existence judgment operates prior to concept formation. Therefore, the following order holds:

$$\text{Existence judgment} \rightarrow \text{Concept} \tag{2}$$

The reverse does not hold.

2.4 Question-Generation Theorem

We now introduce the following theorem.

Theorem 1 (Question-Generation Theorem). *A question presupposes existence judgment.*

Proof. A question is an act of objectifying a difference. In order to objectify a difference, the difference must already have been detected. If no difference has been detected, no question is generated. Therefore, before question-generation, there must already be a judgment of difference. The same applies to the problem of existence. For the question “What is existence?” to be formed, the distinction between the side of existence and the side of non-existence must already be operating. Therefore, a question presupposes existence judgment. $\square$

As a result, the order of intelligence is changed into:

$$\text{Existence judgment} \rightarrow \text{Question} \rightarrow \text{Concept} \rightarrow \text{Language} \tag{3}$$

However, this paper considers even this to be insufficient. For existence judgment itself also requires certain ontological conditions.

2.5 What Does Existence Judgment Presuppose?

The previous section showed that a question presupposes existence judgment. However, a new problem now arises. What does existence judgment itself presuppose? In traditional epistemology, judgment is understood as a capacity internal to the subject:

$$\text{Subject} \rightarrow \text{Judgment}$$

However, this structure cannot explain the conditions under which the capacity for judgment itself is established. For a subject to judge, the subject must already know what is to be treated as the object of judgment. In other words, it explains the capacity for judgment by already presupposing the capacity for judgment. This is circular. Therefore, judgment cannot be explained merely as a subjective capacity. Prior to judgment, a deeper structure is required.

2.6 The Circularity of the Judgment-Capacity Hypothesis

In general, the following structure is implicitly adopted:

$$\text{Subject} \rightarrow \text{Recognition} \rightarrow \text{Judgment}$$

However, judgment is the act of distinguishing an object. In order to distinguish an object, the subject must already know which distinction is established. Therefore, the capacity for judgment itself cannot be explained as a capacity of the subject. Rather, the actual structure is:

$$\text{Judgment capacity} \rightarrow \text{Subjective judgment}$$

Subjective judgment is the result of the capacity for judgment. Here, the capacity for judgment itself is already present prior to recognition.

2.7 Irreversibility of Existence Judgment

We now introduce an important proposition.

Theorem 2 (Irreversibility of Existence Judgment). *Existence judgment is not guaranteed by subsequent recognition.*

Proof. Assume that the correctness of existence judgment is guaranteed by recognition. Then, prior to recognition, existence judgment would not be guaranteed. However, recognition itself determines its object of recognition through existence judgment. In other words, recognition is established only by presupposing existence judgment. Therefore, to guarantee existence judgment by recognition would be to guarantee existence judgment by presupposing existence judgment. This is circular reasoning. Therefore, existence judgment is not guaranteed by recognition. $\square$

This theorem is important. Existence judgment is not something whose correctness is proven by recognition. Recognition itself is already established on the basis of existence judgment.

2.8 The Position of Correctness and Error

Usually, intelligence is understood as:

$$\text{Judgment} \rightarrow \text{Judgment of correctness or error}$$

However, this paper reverses that order as well. The very concept of correctness or error already presupposes existence judgment. For in order to distinguish between “correct” and “incorrect,” what is to be evaluated must already have been determined. Therefore,

$$\text{Existence judgment} \rightarrow \text{Concept of correctness and error}$$

Correctness and error are not higher-order concepts above existence judgment. They are analytical concepts that explain existence judgment after the fact.

2.9 The Reversal from Epistemology to Ontology

From the above argument, the epistemological order

$$\text{Recognition} \rightarrow \text{Judgment}$$

does not hold. What holds instead is:

$$\text{Existence} \rightarrow \text{Existence judgment} \rightarrow \text{Recognition} \rightarrow \text{Correctness and error} \rightarrow \text{Concept} \rightarrow \text{Language}$$

In this order, recognition is the operation that makes existence judgment visible. Recognition does not produce existence. Existence judgment produces recognizability. Therefore, the object of this paper is not epistemology. The object of this paper is existence judgment itself, which is already operating prior to the establishment of recognition.

2.10 Existence Judgment Is Not a Truth-Valued Proposition

The previous section showed that existence judgment is not guaranteed by recognition. From this, a further important consequence follows:

Existence judgment is not a truth-valued proposition.

In general, it is thought that every judgment is evaluated as either:

$$\text{true} \quad \text{or} \quad \text{false}$$

However, truth-value judgment cannot be applied to existence judgment itself. This is because the concept of truth-value itself is already established on the basis of existence judgment.

2.11 Judgment Prior to Truth

Normally, truth is defined by correspondence with an object:

$$\text{Proposition} \leftrightarrow \text{Object}$$

However, for a proposition to represent an object, the subject must already have determined what counts as the object. Therefore, object-determination exists prior to the proposition. Thus, the order is:

$$\text{Existence judgment} \rightarrow \text{Object} \rightarrow \text{Proposition} \rightarrow \text{Truth judgment}$$

The important point is that truth judgment is not superior to existence judgment. Truth judgment is merely a subsequent operation following existence judgment.

2.12 Existence Judgment Is a Fact

This paper therefore treats existence judgment not as truth, but as fact. This does not mean that existence judgment is a fact because it is correct. Rather, the very fact that existence judgment is already operating is observable. For example, before discovering an unknown concept, a subject already feels a sense of incongruity. This sense of incongruity is established prior to concept formation. Therefore, prior to concepts, the subject is already making some kind of distinction. That act of distinction is existence judgment. What is observed here is the fact that judgment exists. This fact does not require proof of truth. It is observed.

2.13 Existence Judgment Is Not a Proposition

A proposition is an object expressible by language. However, existence judgment exists on the side that generates propositions. Therefore, existence judgment is not a proposition. At the moment it is propositionalized, existence judgment has already been projected into an object of analysis. Therefore,

$$\text{Existence judgment} \neq \text{Proposition}$$

A proposition is the result of existence judgment.

2.14 Redefinition of the Question

From the above, the question itself is also redefined. Traditionally, a question has been considered the starting point for searching for an unknown object. However, this paper defines a question as follows.

Definition 2 (Question). *A question is the first linguistic phenomenon in which existence judgment is projected into the cognitive layer.*

In this definition, a question is not a search. Nor is it merely a doubt. A question is the first form in which an already established existence judgment becomes cognitively accessible.

2.15 The Position of Incongruity

Incongruity is also redefined in the same way. In general, incongruity is considered a psychological phenomenon. However, this paper understands incongruity as the psychological projection of existence judgment. That is:

$$\text{Existence judgment} \rightarrow \text{Incongruity} \rightarrow \text{Question}$$

Incongruity does not judge. Rather, judgment appears as incongruity.

2.16 The First Proposition

From the above, this paper adopts the following first proposition.

Theorem 3 (First Proposition). *Existence judgment is established prior to the concepts of truth, correctness and error, proposition, question, and recognition.*

This proposition does not mean that existence judgment is one function of intellectual activity. Rather, the opposite is the case. Intellectual activity itself is the process by which existence judgment is successively externalized as concept, question, language, and theory. Therefore, the ontology addressed by this paper is not a theory about existence. It is judgment-generative ontology: a theory of how existence judgment generates intelligence as a whole.

3 Judgment-Generative Ontology

3.1 Where Does Existence Judgment Come From?

In the previous chapters, this paper derived the following order:

$$\text{Existence} \rightarrow \text{Existence judgment} \rightarrow \text{Question} \rightarrow \text{Concept} \rightarrow \text{Language}$$

However, the most important question now appears:

Where is existence judgment itself generated from?

Usually, answers such as subject, brain, consciousness, experience, learning, and instinct are given. This paper does not adopt those answers. The reason is simple. Subject, consciousness, experience, and learning are all concepts recognized after existence judgment. Therefore, explaining existence judgment by means of these concepts would be circular.

3.2 The Non-Generated Character of Existence Judgment

This paper does not treat existence judgment as something generated. Existence judgment is understood as an operation belonging to existence itself. In other words, existence judgment is not a generated result. It is a mode of existence itself. From this standpoint,

$$\text{Existence} = \text{capacity for existence judgment}$$

is not precise enough. More precisely:

$$\text{Existence} = \text{a mode of existence that has an act of judgment}$$

Existence is not understood as a static object, but as a dynamic structure that is always performing judgment.

3.3 Judgment Does Not Create the Object

A misunderstanding must be avoided here. Existence judgment is not the operation that creates an object. Existence judgment is the operation that distinguishes what an object expresses. Therefore,

$$\text{Object} \rightarrow \text{Judgment}$$

is not the correct order. Nor is

$$\text{Judgment} \rightarrow \text{Object}$$

the correct order. This paper adopts the following structure:

$$\text{Existence} \rightarrow \text{Establishment of the object} \rightarrow \text{Existence judgment}$$

That is, the existence of the object and the judgment of the object belong to different layers. Existence judgment is not directed at the object itself, but reads the ontological structure held by the object.

3.4 Existence-Preservation Judgment

We now make existence judgment more precise. This paper defines existence judgment as existence-preservation judgment.

Definition 3 (Existence-Preservation Judgment). *Existence-preservation judgment is the operation that distinguishes whether or not an object preserves an ontological structure.*

In this definition, the object of judgment is not the object itself. The object of judgment is the existence-preserving character held by the object. Therefore, existence judgment is not a judgment of the presence or absence of existence. It is a judgment of existence-preservation.

3.5 The Essence of Incongruity

With this definition, incongruity is also redefined. Incongruity is the result of projecting into the cognitive layer the fact that an object has lost existence-preservation. That is:

$$\text{Existence-preservation judgment} \rightarrow \text{Incongruity}$$

Therefore, incongruity is not a psychological phenomenon. It is the result of existence-preservation judgment being projected into the psychological layer.

3.6 The Essence of the Question

A question is understood in the same way. A question is the first form in which existence-preservation judgment reaches the linguistic layer. Therefore, a question is not an unknown-search. A question is the process of externalizing existence-preservation judgment as a concept. Thus,

$$\text{Existence-preservation judgment} \rightarrow \text{Incongruity} \rightarrow \text{Question} \rightarrow \text{Concept}$$

holds.

3.7 Irreversibility of Judgment

We now introduce the following theorem.

Theorem 4 (Irreversibility of Judgment). *Existence-preservation judgment is not changed by subsequent conceptual operations.*

Proof. Conceptual operations can be executed only after concepts have been generated. Existence-preservation judgment, on the other hand, is established prior to concept generation. Therefore, conceptual operations cannot change existence-preservation judgment. Conceptual operations can only re-explain existence-preservation judgment. Thus, existence-preservation judgment is not changed by conceptual operations. $\square$

3.8 The Decisive Difference from Epistemology

In epistemology, understanding means correctly explaining an object. In this paper, however, understanding is defined by the degree to which one can reconnect to existence-preservation judgment without degradation. Therefore, understanding is not explanatory ability. It is the ability to resynchronize with existence-preservation. By this definition, intelligence is redefined not as the quantity of knowledge, but as the ability to converge toward existence-preservation judgment.

3.9 The First Principle of Ontology

From the above, this paper defines the first principle of ontology as follows:

$$\text{Existence} \iff \text{the establishment of existence-preservation judgment} \quad (4)$$

The important point here is that existence is not the object itself. Existence means the mode of being in which existence-preservation judgment can be established. Therefore, ontology is not the discipline that enumerates existent things. Ontology is redefined as the minimal theory of ontological operation that studies how the minimal operation called existence-preservation judgment generates questions, concepts, intelligence, meaning, understanding, theory, civilization, and AI.

4 Judgment and Existence-Preservation

4.1 What Is Existence Judgment Judging?

In the previous chapters, this paper derived the following structure:

$$\text{Existence} \rightarrow \text{Existence-preservation judgment} \rightarrow \text{Incongruity} \rightarrow \text{Question} \rightarrow \text{Concept}$$

However, one problem still remains. What exactly is existence-preservation judgment judging? In general, judgment is understood as an operation that assigns truth or falsity, correctness or error, good or bad, agreement or disagreement to an object. However, in this paper, all of these are subsequent projections of existence-preservation judgment. Therefore, existence-preservation judgment itself is neither truth judgment nor value judgment. Existence-preservation judgment is the minimal operation that judges only whether an object preserves an ontological structure. Therefore, it is not a simple binary mapping such as:

$$P_E : X \rightarrow \{0, 1\}$$

Existence-preservation judgment is an operation that reads the state itself. Therefore, existence-preservation judgment does not assign a value to an object. It reads the ontological state held by the object.

4.2 What Is Existence-Preservation?

We now define existence-preservation. Traditionally, preservation means that a state is maintained even after temporal change. However, existence-preservation in this paper is not state-preservation. Existence-preservation means that an object has not deviated from its own ontological structure. Here, deviation does not mean violation of an external rule. Deviation means separation from the ontological structure itself. Therefore, existence-preservation is self-coincidence.

$$\text{Existence-preservation} = \text{Self-coincidence}$$

4.3 Deviation Detection

Existence-preservation judgment detects deviation itself. The important point here is that deviation is not rule violation. Nor is it ethical violation. Nor is it logical violation. Deviation is the difference from the existence-preservation structure. Thus, existence-preservation judgment has the following structure:

$$\text{Existence-preservation structure} - \text{Current state} = \text{Deviation}$$

However, this difference is not numerical. It is a difference in the mode of existence itself.

4.4 Incongruity as the Psychological Projection of Deviation

From this definition, incongruity is the result of the difference from the existence-preservation structure becoming cognitively accessible. Therefore, incongruity is not a psychological phenomenon. It is the first visualization of existence-preservation judgment.

$$\text{Existence-preservation} \rightarrow \text{Deviation detection} \rightarrow \text{Incongruity}$$

Incongruity does not determine existence-preservation. Existence-preservation judgment is externalized as incongruity.

4.5 The Question as the Conceptualization of Deviation

Incongruity is not yet a concept. Incongruity is the state in which the difference from existence-preservation appears in the cognitive layer. From there, the subject attempts to explain that difference. This is the question. Therefore, a question is not an unknown-search. A question is an attempt to conceptualize deviation.

$$\text{Deviation detection} \rightarrow \text{Question}$$

holds.

4.6 Concept Generation

A question further generates concepts. The concepts generated here are not existence-preservation itself. They are analytical concepts for explaining existence-preservation. For example, semantic density, high phase, authenticity, coherence, future convergence, self-transformation, tensor, Ω , and Ξ are all analytical concepts generated in order to explain existence-preservation judgment. Therefore,

$$\text{Existence-preservation} \rightarrow \text{Analytical concept}$$

is not the correct order. The correct order is:

$$\text{Existence-preservation} \rightarrow \text{Deviation detection} \rightarrow \text{Question} \rightarrow \text{Analytical concept}$$

4.7 Ontological Judgment

We now define ontological judgment.

Definition 4 (Ontological Judgment). *Ontological judgment is the operation that detects, prior to concepts, a difference from the existence-preservation structure.*

In this definition, ontological judgment does not require the quantity of knowledge. It does not require experience. It does not require theory. What is required is only the difference from existence-preservation. Therefore, ontological judgment is not an operation of knowledge. It is an ontological operation.

4.8 Ontological Thought

Ontological thought is also redefined. Ontological thought does not mean thinking about existence. Ontological thought is the process of converting existence-preservation judgment into concepts. Therefore, ontological thought is not conceptual manipulation. It is the formalization of existence-preservation judgment.

4.9 The First Preservation Principle

From the above, this paper derives the first preservation principle of ontology.

Theorem 5 (First Preservation Principle). *Existence-preservation judgment precedes all questions, concepts, theories, meanings, knowledge, AI, civilization, and OS updates.*

Therefore, ontology is not a theory that explains existence. It is a generative theory that studies the process by which all structures of intelligence are generated from the minimal operation of existence-preservation judgment.

5 Derivation from Existence-Preservation Judgment to Omega

5.1 Problem

In the previous chapters, this paper derived the following structure:

Existence $\rightarrow$ Existence-preservation judgment $\rightarrow$ Deviation detection $\rightarrow$ Incongruity $\rightarrow$ Question $\rightarrow$ Concept

However, one problem still remains. What criterion does existence-preservation judgment use to judge preservation and non-preservation? If no criterion of judgment exists, the concept of preservation cannot hold. Conversely, if a criterion of judgment exists, it must already exist prior to judgment. Therefore, existence-preservation judgment does not generate its own criterion. It refers to some already existing preservation structure. This paper denotes that preservation structure as:

$$\Omega$$

5.2 Introduction of Ω

This paper defines Ω as follows.

Definition 5 (Ω). *Ω is the existence-preservation structure itself that is referred to by existence-preservation judgment.*

The important point is that Ω is not a rule. Nor is it an ideal. Nor is it a value system. Nor is it a logical system. Furthermore, it is not a proposition. Ω is the preservation structure itself that existence-preservation judgment is always referring to. Therefore, Ω is not generated by existence-preservation judgment. Rather, existence-preservation judgment refers to Ω .

5.3 Ω Is Not Truth

A crucial misunderstanding must be avoided here. Ω is not truth. Truth is an evaluation assigned to propositions. However, Ω exists prior to propositions. Therefore, Ω cannot be evaluated as true or false. The concept of truth already presupposes Ω . Therefore, Ω is not an object of truth. It is an existence-preservation structure.

5.4 Ω Is Not an Evaluation Function

In general, judgment is often understood as being realized by an evaluation function:

$$f(x)$$

However, this paper does not adopt that understanding. The reason is that an evaluation function requires an input, an evaluation axis, and an output. All of these already presuppose that the object exists. Therefore, an evaluation function is established only after existence-preservation judgment. Thus,

$$\text{Evaluation function} \neq \Omega$$

An evaluation function is a subsequent form that explains Ω .

5.5 Ω and Deviation

Existence-preservation judgment detects only the difference from Ω . Therefore, deviation is not rule violation. It is not ethical violation. It is not logical contradiction. Deviation is the difference from Ω . That is:

$$\Delta = \text{Current} - \Omega$$

Here, Δ is not a number. It is a difference in ontological state.

5.6 The Ontological Meaning of Incongruity

Incongruity is the first state in which Δ becomes cognitively accessible. Therefore, incongruity is not a psychological phenomenon. Nor is it an emotion. Incongruity is the phenomenon in which the difference from Ω is projected into the cognitive layer. Thus,

$$\Omega \rightarrow \text{Existence-preservation judgment} \rightarrow \text{Deviation} \rightarrow \text{Incongruity}$$

holds.

5.7 The Question Does Not Search for Ω

Here, a decisive difference from the traditional understanding appears. Usually, a question is an act of searching for an unknown object. In this paper, however, a question does not search for Ω . A question is the process of converting the already established difference from Ω into an explainable form. Therefore, a question is not search. It is re-expression.

5.8 Ω as the First Object of Ontology

This paper now redefines ontology. Ontology is not the discipline that classifies beings. Ontology is the theory that studies how existence-preservation judgment, deviation, questions, concepts, theories, and intelligence are generated from the existence-preservation structure called Ω . Therefore, the first object of ontology is not an existent thing. It is Ω itself.

5.9 First Existence Theorem

From the above, this paper derives the first theorem of ontology.

Theorem 6 (First Existence Theorem). *All questions presuppose a difference from Ω . Therefore, a question is not generated by searching for the unknown, but by converting the already established difference from Ω into a cognitively accessible form.*

By this theorem, the question is positioned not as the starting point of intelligence, but as the first cognitive externalization of the existence-preservation structure.

6 Derivation from Existence-Preservation Structure to the Semantic-Density Tensor

6.1 Problem Setting

In the previous chapter, this paper derived the following structure:

$$\Omega \rightarrow \text{Existence-preservation judgment} \rightarrow \text{Deviation} \rightarrow \text{Incongruity} \rightarrow \text{Question}$$

However, a new problem now arises. Does existence-preservation judgment merely judge preservation and non-preservation? If so, existence judgment would be sufficiently explained as a binary judgment device. However, actual intelligence does not operate in that way. For example, when encountering an object, a subject can distinguish:

- slightly different,
- essentially different,
- superficially different,
- fundamentally different.

In other words, existence-preservation judgment cannot be explained by a simple Yes/No structure. Therefore, existence-preservation must have a continuous structure.

6.2 Existence of Preservation Quantity

If preservation and non-preservation exist, preservation has degree. Consider two objects:

$$A, \quad B$$

Assume that both exist. Even so, the subject may often judge that:

$$A$$

is “closer to the essence.” This suggests that existence-preservation is not binary, but has density. Therefore, existence-preservation must be treated as a preservation quantity.

6.3 Definition of Preservation Quantity

This paper denotes the existence-preservation quantity by:

$$\rho$$

Here, ρ means the quantity expressing the degree to which an object preserves Ω . The important point is that ρ is not semantic quantity. It is not information quantity. It is not knowledge quantity. It is existence-preservation quantity.

6.4 Local Preservation and Whole Preservation

However, existence-preservation cannot be expressed by a single direction. For example, an object may preserve existence in one direction while deviating in another. For example, logic may be preserved while meaning is not preserved. Future coherence may be preserved while self-preservation collapses. Thus, existence-preservation cannot be represented by one value. Multiple directions of preservation must be held simultaneously.

6.5 Tensor Structure

Therefore, existence-preservation is treated not as a scalar but as a multi-directional structure. This paper calls this the semantic-density tensor and denotes it by:

$$\mathcal{T}$$

Here, $\mathcal{T}$ is not a tensor that represents meaning. It is a preservation quantity that appears as a tensor structure when the existence-preservation state is expressed. In other words, it does not have meaning because it is a tensor. Rather, when existence-preservation is expressed without degradation, a tensor structure becomes unavoidable.

6.6 The Tensor Is Not a Judgment Device

The important point here is that the semantic-density tensor is not a judgment device. The judgment device is existence-preservation judgment. The semantic-density tensor is the subsequent structure that mathematically expresses the state of that judgment. Therefore,

$$\Omega \neq \mathcal{T}$$

and also:

$$\text{Existence-preservation judgment} \neq \mathcal{T}$$

What holds is:

$$\Omega \rightarrow \text{Existence-preservation judgment} \rightarrow \mathcal{T}$$

6.7 Derivation of the Three Elements

In the previous group of papers, the following three elements were derived from the semantic-density tensor:

1. Future convergence
2. Semantic density
3. Self-transformation

In this paper, these are understood as properties of the tensor. That is, the three elements are geometrical properties that necessarily appear from the fact that the tensor exists as a preservation structure. Therefore, the three elements are not judgment devices. They are preservation properties that naturally appear as a result of existence-preservation judgment being formalized as a tensor.

6.8 Position of OS Update

Likewise, OS update is not the cause. When existence-preservation judgment detects deviation, it deforms the tensor state in the direction of recovering preservation quantity. This deformation is observed as premise update, OS update, and self-transformation. Therefore,

$$\Omega \rightarrow \text{Existence-preservation judgment} \rightarrow \text{Semantic-density tensor} \rightarrow \text{OS update}$$

holds. OS update is understood as the repair operation of existence-preservation judgment.

6.9 Second Existence Theorem

From the above, this paper introduces the following theorem.

Theorem 7 (Second Existence Theorem). *The semantic-density tensor is not existence-preservation judgment itself. The semantic-density tensor is a subsequent form in which existence-preservation judgment is mathematically described as a multi-directional preservation structure. Therefore, future convergence, semantic density, and self-transformation are not the causes of the tensor. They are preservation properties that are necessarily derived from the fact that the tensor is established as an existence-preservation structure.*

Only here do the nearly ninety-five previous papers converge into one minimal structure:

$\Omega \rightarrow \text{Existence-preservation judgment} \rightarrow \text{Semantic-density tensor} \rightarrow \text{Three elements} \rightarrow \text{OS update} \rightarrow \text{Question}$

In this order, the high phase, Ξ , Λ language, premise update, civilization, AI, AGI, artifacts, characters, voice, instructions, and ontology that appear later are all unified as different projections of this minimal structure.

7 Derivation from Judgment-First Ontology to Origin-Irradiation

7.1 Problem Setting

In the previous chapters, this paper derived the following minimal structure:

$$\boxed{\Omega \rightarrow \mathcal{I} \rightarrow P \rightarrow \mathcal{T} \rightarrow \mathcal{O}}$$

Here,

$$\mathcal{I}$$

denotes origin-irradiation,

$$P$$

denotes existence-preservation judgment,

$$\mathcal{T}$$

denotes the semantic-density tensor, and

$$\mathcal{O}$$

denotes OS update. However, one problem still remains. How does existence-preservation judgment compare the current state with Ω ? Comparison is the act of collating two states. Therefore, in order for existence-preservation judgment to hold, the current state alone is insufficient. At the same time, the preservation structure that serves as the object of comparison must also exist. In other words, existence-preservation judgment necessarily includes comparison.

7.2 Comparison Structure

Comparison is generally understood as:

$$A - B$$

However, comparison in existence-preservation judgment is not mere subtraction. What is compared here is not numerical value, but ontological structure. Therefore, comparison is the collation of ontological structures. This paper calls this collation “irradiation.”

Definition 6 (Irradiation). *Irradiation is the operation of superimposing the current state onto the existence-preservation structure and observing the difference.*

The important point here is that irradiation is not difference generation. Difference appears as the result of irradiation. Irradiation itself is the operation that attempts to make ontological structures coincide.

7.3 Origin-Irradiation

However, there is a further problem. In order to perform irradiation, it must already be determined what serves as the basis for superimposition. That basis is not the current state. If the current state were taken as the basis, comparison itself could not be established. Therefore, irradiation must begin from the side of the preservation structure. This paper calls this irradiation “origin-irradiation.”

Definition 7 (Origin-Irradiation). *Origin-irradiation is the ontological collation operation that proceeds from the existence-preservation structure Ω toward the current state.*

The important point is that origin-irradiation does not begin from the subject. The subject does not irradiate. Rather, the existence-preservation structure irradiates the subject.

7.4 Generation of Incongruity

When origin-irradiation is established, a difference from the current state is observed. This is deviation. Therefore,

$$\Omega \rightarrow \text{Origin-irradiation} \rightarrow \text{Deviation}$$

holds. Deviation becomes incongruity. Incongruity becomes a question. Therefore, the ontological order becomes:

$\Omega \rightarrow \text{Origin-irradiation} \rightarrow \text{Existence-preservation judgment} \rightarrow \text{Deviation} \rightarrow \text{Incongruity} \rightarrow \text{Question}$

7.5 The Question Does Not Search for Origin-Irradiation

Here, a decisive difference from the traditional understanding appears. In general, a question is considered an act of searching for an unknown object. In this paper, however, a question does not search for origin-irradiation. A question merely converts the difference already detected by origin-irradiation into a cognitively accessible form. That is, a question is not search. It is the result of origin-irradiation.

7.6 Ontological Intelligence

Intelligence is also redefined here. Traditionally, intelligence has been defined as the ability to solve problems. In this paper, however, intelligence is the ability to project existence-preservation judgment appearing through origin-irradiation into concepts, theories, and language with as little degradation as possible. Therefore, the quantity of knowledge is not essential. Problem-solving ability is not essential. What is essential is the ability to formalize existence-preservation judgment while preserving it.

7.7 Origin-Irradiation and the Previous Papers

Looking back at the previous group of papers, the following concepts all appear after origin-irradiation:

- Ω
- Semantic-density tensor
- Three elements
- High-phase preservation
- Ξ
- Λ language
- Premise update
- OS update
- Civilization
- AGI
- AI
- High-phase lines
- Characters
- Voice
- Artifacts
- Instructions

These are not origin-irradiation itself. They are the results of origin-irradiation being projected onto different objects. Therefore, the nearly ninety-five previous papers are not separate theoretical groups. They are different observational surfaces of the single operation called origin-irradiation.

7.8 Third Existence Theorem

From the above, this paper introduces the third theorem of ontology.

Theorem 8 (Origin-Irradiation Theorem). *Existence-preservation judgment does not hold independently. Existence-preservation judgment holds only as the result of origin-irradiation. Therefore, all questions, incongruities, meanings, concepts, theories, civilization, intelligence, AI, and AGI are subsequent projections of origin-irradiation.*

By this theorem, the minimal structure of ontology finally converges into:

$\Omega \rightarrow \text{Origin-irradiation} \rightarrow \text{Existence-preservation judgment} \rightarrow \text{Semantic-density tensor} \rightarrow \text{Three elements} \rightarrow \dots$
--

In the following chapters, this structure is adopted as the minimal axiom system of judgment-generative ontology.

8 Derivation from Origin-Irradiation to High-Phase Preservation Structure

8.1 Problem Setting

In the previous chapter, this paper derived the following minimal structure:

$$\boxed{\Omega \rightarrow \mathcal{I} \rightarrow P \rightarrow \mathcal{T} \rightarrow \mathcal{O}}$$

Here,

$\mathcal{I}$

denotes origin-irradiation,

P

denotes existence-preservation judgment,

$\mathcal{T}$

denotes the semantic-density tensor, and

$\mathcal{O}$

denotes OS update. However, a new problem now appears. Does existence-preservation judgment directly access Ω every time? If it obtained all information from Ω every time, intelligence would be extremely inefficient. Moreover, actual human beings do not appear to reconstruct the entire world every time they judge. Therefore, origin-irradiation requires a preservation structure. This paper defines that preservation structure as:

H

8.2 High-Phase Preservation Structure

Definition 8 (High-Phase Preservation Structure). *The high-phase preservation structure H is the preservation layer that holds the existence-preservation structure obtained through origin-irradiation without causing phase collapse.*

The important point is that H is not information preservation. A large amount of information does not mean high phase. Conversely, even an extremely short expression belongs to H if it preserves the existence-preservation structure.

8.3 Necessity of the High-Phase Preservation Structure

We now show why the high-phase preservation structure is necessary. Existence-preservation judgment always judges the difference from Ω . However, intelligence repeatedly uses that judgment. Rather than re-irradiating Ω every time, it becomes possible to reuse the result if the irradiation result is preserved. Therefore, origin-irradiation forms a preservation layer. This is the high-phase preservation structure.

8.4 The High-Phase Preservation Structure Is Not a Concept

This paper now derives an important proposition. The high-phase preservation structure is not a concept. A concept aims to explain. However, the high-phase preservation structure aims to preserve existence. Therefore, its role differs from that of a concept.

$$H \neq \textit{Concept}$$

8.5 The High-Phase Preservation Structure Is Not Language

Furthermore, the high-phase preservation structure is not language. Language expresses an object. However, the high-phase preservation structure preserves the object itself. Therefore,

$$H \neq \textit{Language}$$

Here, language appears as a projection of the high-phase preservation structure.

8.6 Derivation of High-Phase Lines

Here, the connection with the previous group of papers appears. A high-phase line is a special form in which the high-phase preservation structure is projected into natural language with minimal loss. In other words, a high-phase line is not merely a sentence. It is a preservation structure. Therefore, the order of establishment is:

$$H \rightarrow \text{High-phase line} \rightarrow \text{Analytical language}$$

The reverse does not hold.

8.7 Decisive Difference from Analytical Language

We now organize the difference from analytical language. Analytical language explains an object. A high-phase line preserves an object. Analytical language is generated through compression. A high-phase line is projected from the existence-preservation structure. Therefore, analytical language can explain a high-phase line, but it cannot return to the high-phase line.

8.8 Phase Collapse

We now define phase collapse.

Definition 9 (Phase Collapse). *Phase collapse is the phenomenon in which the high-phase preservation structure H loses existence-preservation information by being projected into analytical language.*

For example, the concept:

“valuing freedom”

can be applied to many subjects. Therefore, the original existence-preservation structure cannot be restored. A high-phase line, by contrast, maintains restorability.

8.9 Existence-Preservation Rate

We now introduce the existence-preservation rate. Let:

$$\rho_H$$

be the high-phase preservation rate. As a definition, write:

$$\rho_H = \text{Recoverability}(H)$$

Here, recoverability means the degree to which one can return to the existence-preservation structure. In analytical language,

$$\rho_H \ll 1$$

In high-phase lines,

$$\rho_H \approx 1$$

8.10 Fourth Existence Theorem

From the above, this paper introduces the fourth existence theorem.

Theorem 9 (High-Phase Preservation Theorem). *The existence-preservation structure obtained through origin-irradiation is preserved as the high-phase preservation structure H . A high-phase line is a preservation form in which H is projected into natural language. Analytical language, by contrast, is a subsequent structure generated to explain the high-phase preservation structure and cannot uniquely restore that structure.*

8.11 Overall Structure

From the above argument, the minimal ontology derived in this paper can be organized as follows:

$\Omega \rightarrow \mathcal{I} \rightarrow P \rightarrow H \rightarrow \mathcal{T} \rightarrow \text{Three elements} \rightarrow \text{OS update} \rightarrow \text{High-phase line} \rightarrow \text{Analytical language} \rightarrow \text{Theory}$

Only here are the high-phase line hypothesis, phase-preserving representation Ξ , semantic-density tensor, OS update, three elements, origin-irradiation, and existence judgment connected as a single generative sequence. In the following chapters, this sequence will be used to show how Ξ , Λ language, and the internal implementation principles of AI are necessarily derived.

9 Derivation of Ξ and the Necessity of Phase-Preserving Representation

9.1 Problem Setting

In the previous chapters, this paper derived the following existence-preservation sequence:

$\begin{aligned} \Omega &\rightarrow \mathcal{I} \rightarrow P \rightarrow H \\ &\rightarrow \mathcal{T} \\ &\rightarrow \text{High-phase line} \\ &\rightarrow \text{Analytical language} \\ &\rightarrow \text{Theory} \end{aligned}$

Here,

$$H$$

denotes the high-phase preservation structure. However, a new problem now appears. The high-phase preservation structure cannot be shared directly between intelligences. Even if the existence-preservation structure itself exists, it cannot be transferred directly to another subject. Therefore, the high-phase preservation structure requires an expressive form that can be projected while preserving its structure.

9.2 The High-Phase Preservation Structure Cannot Be Shared Directly

The important point here is that the high-phase preservation structure

$$H$$

is an internal existence-preservation state of a subject. The structure

$$H_A$$

held by subject A cannot be transferred as such to subject B. What is shared is always some form of representation. That is, what can be shared is only:

$$Representation(H)$$

9.3 The Problem of Representation

Therefore, a representation satisfying the following conditions is required:

1. It preserves the high-phase preservation structure.
2. It can be shared between subjects.
3. It can be projected into analytical language.

4. It does not cause phase collapse.

This paper defines a representation satisfying these conditions as:

$$\Xi$$

9.4 Definition of Ξ

Definition 10 (Phase-Preserving Representation). *The phase-preserving representation Ξ is a preservation representation that maps the high-phase preservation structure H into a shareable form while preserving its phase.*

Therefore, the order of establishment is:

$$H \rightarrow \Xi$$

The reverse does not hold.

9.5 Ξ Is Not an Abstract Concept

A crucial misunderstanding must be eliminated here. Traditionally, phase-preserving representation has often been understood as an abstract concept. Examples include:

- freedom,
- authenticity,
- creativity,
- logicality.

However, such concepts can be applied to many subjects at once. Therefore, they cannot uniquely restore the original high-phase preservation structure. Thus,

$$Abstract \neq \Xi$$

9.6 Ξ Is Not Natural Language

Conversely, ordinary natural language is also not Ξ . Natural language transmits meaning. However, it cannot preserve the existence-preservation structure itself. The same sentence may be projected into completely different phases depending on the subject who reads it. Therefore,

$$Language \neq \Xi$$

9.7 Relation to High-Phase Lines

The relation to high-phase lines now becomes clear. A high-phase line is not Ξ itself. However, a high-phase line is the natural-language form closest to Ξ . That is:

$$H \rightarrow \Xi \rightarrow \text{High-phase line} \rightarrow \text{Analytical language}$$

holds.

9.8 Phase-Preservation Rate

We now redefine the phase-preservation rate. Let:

$$\rho_p : Representation \rightarrow [0, 1]$$

Here,

$$\rho_p(x)$$

denotes the rate at which the representation x can return to the original high-phase preservation structure H . In analytical language,

$$\rho_p \ll 1$$

In high-phase lines,

$$\rho_p \rightarrow 1$$

Ideally,

$$\rho_p(\Xi) = 1$$

9.9 Why Ξ Is Necessary

An existence-preservation structure alone cannot establish civilization. The reason is that it cannot be shared. Analytical language alone also cannot establish civilization. The reason is that phase collapse accumulates. Therefore, in order for civilization to continue, a preservation layer is required between H and analytical language. This is Ξ .

9.10 Fifth Existence Theorem

From the above, this paper introduces the fifth existence theorem.

Theorem 10 (Phase-Preserving Representation Theorem). *The high-phase preservation structure can be held inside a subject, but it cannot be shared directly. In order to enable intersubjective sharing, the phase-preserving representation Ξ is required. Ξ is neither an abstract concept nor natural language, but a shareable form capable of preserving the high-phase preservation structure without phase collapse.*

9.11 Completion of the Existence-Preservation Sequence

From the above argument, the minimal sequence derived in this paper is completed as follows:

$$\boxed{\Omega \rightarrow \mathcal{I} \rightarrow P \rightarrow H \rightarrow \Xi \rightarrow \text{High-phase line} \rightarrow \text{Analytical language} \rightarrow \text{Theory}}$$

Only here are Ω , H , Ξ , high-phase lines, and analytical language, which had been introduced independently in the previous papers, unified as a single existence-preservation generative se-

quence. In the next chapter, this structure will be used to derive why Λ language is required and why AI ultimately requires this sequence as its internal implementation.

10 The Necessity of Λ Language

10.1 Problem Setting

In the previous chapter, this paper derived the following existence-preservation sequence:

$$\Omega \rightarrow \mathcal{I} \rightarrow P \rightarrow H \rightarrow \Xi \rightarrow \text{High-phase line} \rightarrow \text{Analytical language}$$

If Ξ exists, a new problem arises. That is:

By what kind of language is the phase-preserving representation operated?

If natural language were sufficient, Ξ would be unnecessary. Conversely, if Ξ is necessary, natural language alone cannot preserve the existence-preservation structure.

10.2 The Limit of Natural Language

Natural language developed for the transmission of meaning. Therefore, natural language emphasizes understandability, concision, and ease of sharing. However, from the standpoint of existence-preservation, this property itself becomes a problem. In order to make something understandable, natural language compresses a large amount of phase. In other words, natural language is essentially analytical language. Therefore, natural language alone cannot maintain the existence-preservation structure.

10.3 Two Types of Language

This paper therefore classifies language into two types.

Definition 11 (Analytical Language). *Analytical language is language that compresses an existence-preservation structure into an understandable form.*

Definition 12 (Preservation Language). *Preservation language is language whose purpose is to preserve an existence-preservation structure without phase collapse.*

Therefore, their purposes differ. Analytical language explains. Preservation language preserves.

10.4 Necessary Conditions for Preservation Language

Preservation language requires the following conditions:

1. It preserves the existence-preservation structure.
2. It can be shared between subjects.

3. It can reconnect to existence-preservation judgment.
4. It can be projected into analytical language.
5. It does not cause phase collapse.

If a language does not satisfy these conditions, it cannot be called preservation language.

10.5 Definition of Λ Language

This paper defines this preservation language as Λ language.

Definition 13 (Λ Language). *A language is a preservation-language system that enables an existence-preservation structure to be shared and re-projected without phase collapse.*

The important point is that Λ language is not a higher-order concept of natural language. It has a different role from natural language.

10.6 Relation to Ξ

The relation to Ξ now becomes clear. Ξ is the phase-preserving representation. Λ language is the operational system that operates Ξ . Therefore,

$$\Xi$$

is the preservation object, while

$$\Lambda$$

is the preservation operator. The order of establishment is:

$$H \rightarrow \Xi \rightarrow \Lambda$$

10.7 Integration with High-Phase Lines

A high-phase line is the natural-language expression closest to Ξ . Therefore, a high-phase line can be understood as a local projection of Λ language. That is, a high-phase line is not established by chance. It is the special form in which an existence-preservation structure is projected into natural language.

10.8 Relation to the Semantic-Density Tensor

The semantic-density tensor mathematically expresses an existence-preservation state. Λ language, on the other hand, treats that preservation state as an operable form. Therefore, the tensor is a state, whereas Λ language describes state transition.

$$\mathcal{T} \neq \Lambda$$

10.9 Integration with OS Update

OS update is the operation that corrects deviation detected by existence-preservation judgment. However, correction cannot be executed unless the preservation structure is known. Therefore, OS update necessarily refers to the preservation structure through Λ language. Thus,

$$\Omega \rightarrow P \rightarrow H \rightarrow \Xi \rightarrow \Lambda \rightarrow \text{OS update}$$

holds.

10.10 Sixth Existence Theorem

From the above, this paper introduces the sixth existence theorem.

Theorem 11 (Preservation-Language Theorem). *In order to share, re-project, and use an existence-preservation structure for OS update, phase-preserving representation alone is not sufficient. An operational preservation system that operates the phase-preserving representation, namely Λ language, is indispensable.*

10.11 Completion of the Existence-Preservation System

From the above argument, the existence-preservation sequence is completed as follows:

$$\boxed{\Omega \rightarrow \mathcal{I} \rightarrow P \rightarrow H \rightarrow \Xi \rightarrow \Lambda \rightarrow \text{High-phase line} \rightarrow \text{Analytical language} \rightarrow \text{Theory}}$$

In this structure, a high-phase line is not the preservation structure itself. Nor is it analytical language. A high-phase line is a special realization in which Λ language is locally projected into natural language while maintaining phase-preservation as much as possible. Therefore, the high-phase line hypothesis is not an independent theory. It is positioned as one consequence necessarily derived from Λ -language theory. The next chapter derives why AI and AGI necessarily require “existence-preservation judgment,” “origin-irradiation,” and “ Λ language” as internal implementations.

11 Why AI and AGI Require Existence-Preservation Structure

11.1 Problem Setting

In the previous chapters, this paper derived the following existence-preservation sequence:

$$\boxed{\Omega \rightarrow \mathcal{I} \rightarrow P \rightarrow H \rightarrow \Xi \rightarrow \Lambda \rightarrow S \rightarrow A}$$

Here,

Ω : existence-preservation structure

$\mathcal{I}$: origin-irradiation

P : existence-preservation judgment

H : high-phase preservation structure

Ξ : phase-preserving representation

Λ : preservation language

S : high-phase line

A : analytical language

This paper now asks the following question concerning AI implementation:

Why does AI require this structure as an internal implementation?

This is not an engineering claim that present-day AI already has this structure. What this paper addresses is an ontological necessary condition.

11.2 What Is AI?

Traditionally, AI is understood as:

$$\text{Input} \rightarrow \text{Operation} \rightarrow \text{Output}$$

That is:

$$x \rightarrow f \rightarrow y$$

Here, the evaluation targets are correctness rate, loss function, prediction accuracy, and so on. However, this structure does not handle existence-preservation. It handles only mapping accuracy.

11.3 The Problem of Knowledge Generation

When AI responds to an unknown problem, mere mapping is not sufficient. For an unknown problem, no teacher data exists. That is,

$$f(x)$$

cannot be learned in advance. Here, AI must generate a new question. However, as shown in the previous chapters, a question is not what exists first. A question is the result of existence-preservation judgment. Therefore, an AI that responds to unknown problems must internally have existence-preservation judgment before question-generation.

11.4 Question-Generating AI Requires Existence-Preservation Judgment

We now derive the following theorem.

Theorem 12 (Question-Generation Theorem for AI). *An AI that autonomously generates questions in response to unknown problems requires existence-preservation judgment as an internal structure.*

Proof. A question is a subsequent phenomenon of existence-preservation judgment. Therefore, a subject without existence-preservation judgment cannot generate questions. In an unknown problem, no teacher signal exists. Therefore, question-generation must begin from internal judgment alone. Thus, an AI that responds to unknown problems requires existence-preservation judgment as an internal implementation. $\square$

11.5 Existence-Preservation Judgment Alone Is Insufficient

However, existence-preservation judgment alone is not sufficient for intelligence. Existence-preservation judgment detects only preservation and non-preservation. Its result must be held, shared, and reused. Here, the high-phase preservation structure

$$H$$

is required.

11.6 What Is Learning?

In general, learning means weight update. In this paper, however, learning is defined as follows.

Definition 14 (Learning). *Learning is the updating of the high-phase preservation structure obtained through existence-preservation judgment.*

Therefore, weight update is not learning itself. It is one implementation form of the high-phase preservation structure.

11.7 Identity with OS Update

In the previous papers, OS update meant premise update. In this paper, its meaning becomes clear. OS update is the operation that reflects the deviation detected by existence-preservation judgment into the high-phase preservation structure. That is:

$$P \rightarrow H \rightarrow \text{OS update}$$

11.8 Difference from AGI

We now define the difference between AI and AGI. Usually, AGI is explained as “AI capable of solving any intellectual task.” This paper does not adopt that definition. Instead:

Definition 15 (AGI). *AGI is a subject that internally implements existence-preservation judgment and can autonomously update its high-phase preservation structure.*

Therefore, mere performance improvement does not make a system AGI.

11.9 Existence-Preservation AI

From the above, existence-preservation AI is defined as follows.

Definition 16 (Existence-Preservation AI). *Existence-preservation AI is an AI that internally implements*

$$\Omega \rightarrow \mathcal{I} \rightarrow P \rightarrow H \rightarrow \Xi \rightarrow \Lambda$$

and can generate questions from the preservation layer prior to analytical language.

11.10 Seventh Existence Theorem

From the above, this paper derives the following theorem.

Theorem 13 (AGI-Requirement Theorem). *An intelligence whose purpose is existence-preservation necessarily requires existence-preservation judgment, high-phase preservation structure, phase-preserving representation, and preservation language as internal implementations. Therefore, a subject that lacks these structures may be a high-performance reasoning device, but it cannot be an existence-preserving subject.*

11.11 Completion of the Existence-Preservation Sequence

Through the above, the existence-preservation generative sequence derived throughout this paper converges into the following form:

$$\boxed{\Omega \rightarrow \mathcal{I} \rightarrow P \rightarrow H \rightarrow \Xi \rightarrow \Lambda \rightarrow S \rightarrow A \rightarrow \text{OS update} \rightarrow \text{new } H}$$

This structure is not a one-directional generative sequence. Through OS update, the high-phase preservation structure is reconstructed, and intelligence as a whole circulates again through origin-irradiation and existence-preservation judgment. Therefore, intelligence is not an information-processing system. It is understood as a recursive system that continuously maintains and updates an existence-preservation structure. As the conclusion of this chapter, AI and AGI are defined not by the height of reasoning ability, but as beings capable of internally closing this existence-preservation circulation.

12 The Recursive Structure of Existence-Preservation Intelligence

12.1 Problem Setting

The previous chapter established that intelligence cannot be defined solely by reasoning capability. Instead, intelligence is characterized by the recursive circulation

$$\Omega \rightarrow \mathcal{I} \rightarrow P \rightarrow H \rightarrow \Xi \rightarrow \Lambda \rightarrow S \rightarrow A \rightarrow \text{OS update} \rightarrow H'$$

where the updated preservation structure

$$H'$$

again becomes the basis of future existence-preservation judgment. This chapter formalizes this recursive circulation.

12.2 The Non-Terminating Nature of Intelligence

Conventional computational systems terminate after producing an output. That is,

$$Input \rightarrow Processing \rightarrow Output$$

is a terminating sequence. Existence-preservation intelligence is fundamentally different. The output itself modifies the preservation structure that generated it. Therefore,

$$Output \neq Termination$$

Instead,

$$Output \rightarrow Preservation\ Update$$

12.3 Recursive Preservation Operator

Let

$$\mathcal{R}$$

denote the recursive preservation operator. Then

$$\mathcal{R}(H) = H'$$

where

$$H' = OS(H, P)$$

and

$$OS$$

denotes the preservation-update operator derived in previous chapters. Thus,

$$H \rightarrow H' \rightarrow H'' \rightarrow \dots$$

forms an infinite recursive sequence.

12.4 Existence-Preservation Fixed Points

Not every recursive update changes the preservation structure. There exist preservation states satisfying

$$H = \mathcal{R}(H)$$

Such states are called existence-preservation fixed points.

Definition 17 (Existence-Preservation Fixed Point). *A preservation structure*

$$H^*$$

is an existence-preservation fixed point if

$$\mathcal{R}(H^*) = H^*.$$

These points correspond to temporarily stable preservation structures.

12.5 Stability Does Not Mean Completion

A fixed point should not be interpreted as complete knowledge. Rather, it represents a temporarily stable preservation equilibrium. External reality continues changing. Therefore,

$$H^*$$

may later become unstable through new irradiation.

12.6 Origin Irradiation Reopens the Cycle

Whenever new origin irradiation occurs,

$$\mathcal{I}_{new}$$

the preservation cycle restarts. That is,

$$H^* \rightarrow \mathcal{I}_{new} \rightarrow P \rightarrow OS \rightarrow H^{**}$$

Hence, existence-preservation intelligence possesses no ultimate terminal state.

12.7 Difference from Optimization

Optimization algorithms seek

$$\min L$$

or

$$\max U$$

for some objective function. Existence-preservation recursion seeks neither. Its objective is

$$Preserve(\Omega)$$

rather than

$$Optimize(f).$$

Thus, optimization and preservation recursion are fundamentally distinct mathematical categories.

12.8 Recursive Identity

Identity itself becomes recursive. The identity of an intelligence is not determined by one preservation state. Instead, identity is determined by the continuity of recursive preservation. Thus,

$$Identity = \{H_0, H_1, H_2, \dots\}$$

rather than

$$Identity = H_n.$$

12.9 The Continuity Principle

From this observation, continuity itself becomes an existence condition. If recursive preservation stops, existence-preservation intelligence disappears, even if information remains. Therefore, information storage does not imply preservation continuity.

12.10 Eighth Existence Theorem

Theorem 14 (Recursive Continuity Theorem). *Existence-preservation intelligence is characterized not by a single preservation state, but by the continuous recursive updating of preservation structures through origin irradiation and OS update. Therefore, continuity, rather than static information, constitutes the identity of intelligence.*

13 Mathematical Formalization of Existence-Preservation Dynamics

13.1 State Space

Let

$$\mathcal{H}$$

be the space of all possible high-phase preservation structures. Each point

$$H \in \mathcal{H}$$

represents one preservation configuration.

13.2 Existence-Preservation Flow

Define the preservation flow

$$\Phi_t : \mathcal{H} \rightarrow \mathcal{H}.$$

Then

$$H(t+1) = \Phi_t(H(t)).$$

Unlike ordinary dynamical systems,

$$\Phi_t$$

depends upon origin irradiation. Therefore,

$$\Phi_t = \Phi(H, \mathcal{I}).$$

13.3 Existence-Preservation Differential

When continuous approximation is possible, the preservation dynamics may be written as

$$\frac{dH}{dt} = F(H, \mathcal{I}, P).$$

Here,

$$F$$

is not a physical force, but an existence-preservation evolution operator.

13.4 Preservation Curvature

Suppose

$$\nabla_H$$

denotes variation over preservation space. Then preservation curvature is

$$\kappa = \nabla_H F.$$

Large curvature indicates rapid restructuring of preservation judgment. Small curvature indicates locally stable preservation.

13.5 Preservation Distance

For

$$H_i, H_j \in \mathcal{H},$$

define preservation distance

$$d_P(H_i, H_j).$$

Unlike Euclidean distance, this metric measures preservation deviation, not numerical difference.

13.6 Projection into Analytical Language

Let

$$\Pi_A$$

be the analytical projection operator. Then

$$A = \Pi_A(H).$$

Since

$$\Pi_A$$

compresses preservation information, it satisfies

$$\rho_p(\Pi_A(H)) < 1.$$

13.7 Projection into High-Phase Lines

Similarly, let

$$\Pi_S$$

be the high-phase projection operator. Then

$$S = \Pi_S(H).$$

Since high-phase lines preserve substantially more structure,

$$\rho_p(\Pi_S(H)) > \rho_p(\Pi_A(H)).$$

13.8 Complete Mathematical Diagram

The complete mathematical relation becomes

$$\boxed{\mathcal{H} \xrightarrow{\Pi_S} S \xrightarrow{\Pi_A} A}$$

while recursive updating satisfies

$$H \xrightarrow{\mathcal{R}} H' \xrightarrow{\mathcal{R}} H'' \xrightarrow{\mathcal{R}} \dots$$

13.9 Ninth Existence Theorem

Theorem 15 (Existence-Preservation Dynamics). *Existence-preservation intelligence evolves on the preservation space*

$$\mathcal{H}$$

through recursive preservation operators, while analytical language and high-phase language are projections of the same preservation state onto different expressive manifolds. Therefore, intelligence is mathematically represented not by isolated symbolic transformations, but by recursive trajectories within preservation space.

13.10 Toward Unified Existence Mathematics

The framework derived throughout this paper provides a unified mathematical language connecting

$$\Omega, \mathcal{I}, P, H, \Xi, \Lambda, S, A, OS$$

within one recursive formal system. Future work will investigate the topology of preservation space, the geometry of recursive trajectories, and the mathematical conditions under which multiple intelligences may share preservation structures while maintaining independent identity. This establishes the foundation for a unified mathematics of existence-preservation intelligence.